

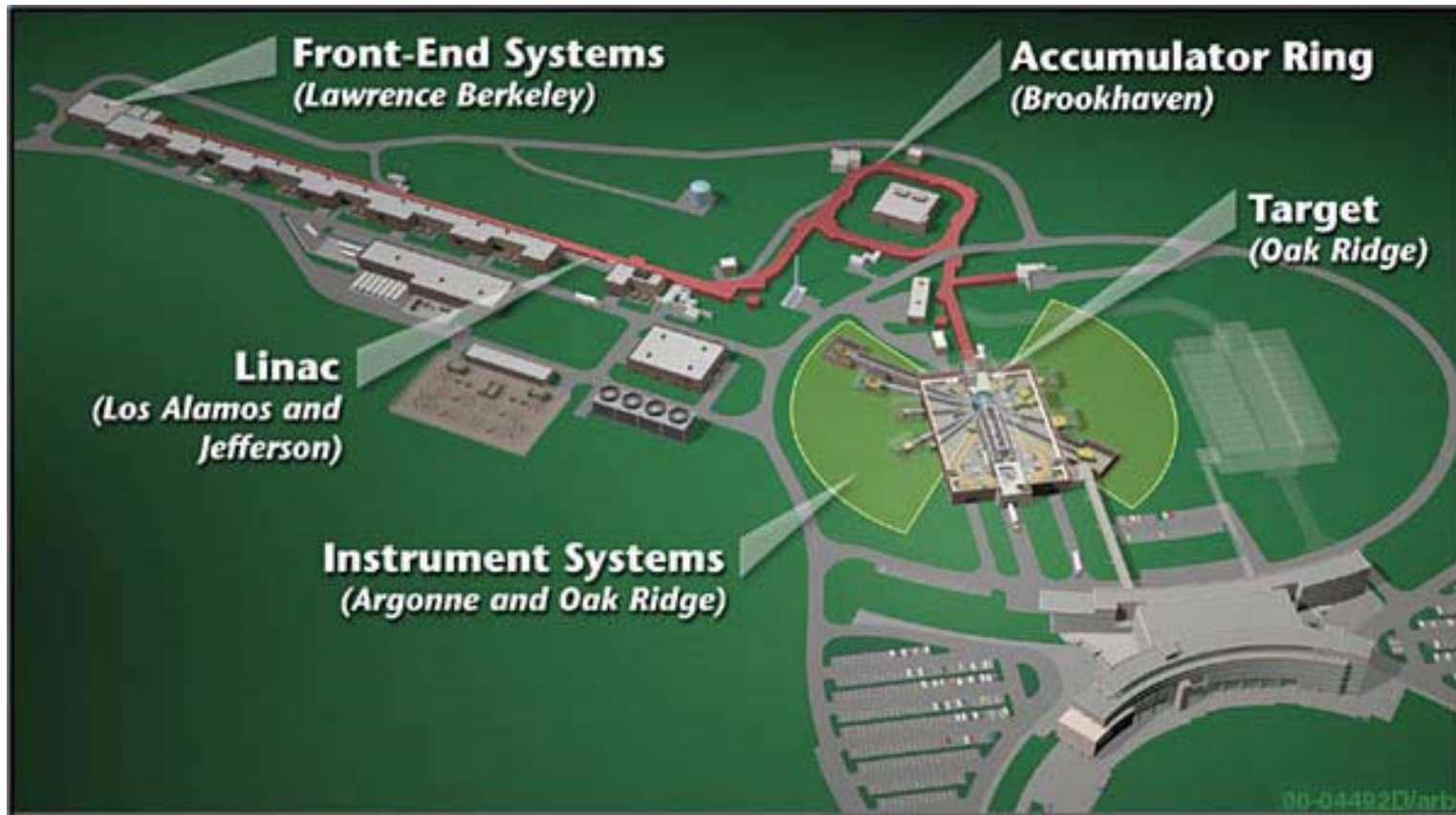
Development and Operation of the SNS Fast Chopper Systems

Robert Saethre



*Work performed at Oak Ridge National Laboratory, which is managed by UT-Battelle, LLC, under contract DE-AC05-00OR22725 for the U.S. Department of Energy.

Spallation Neutron Source



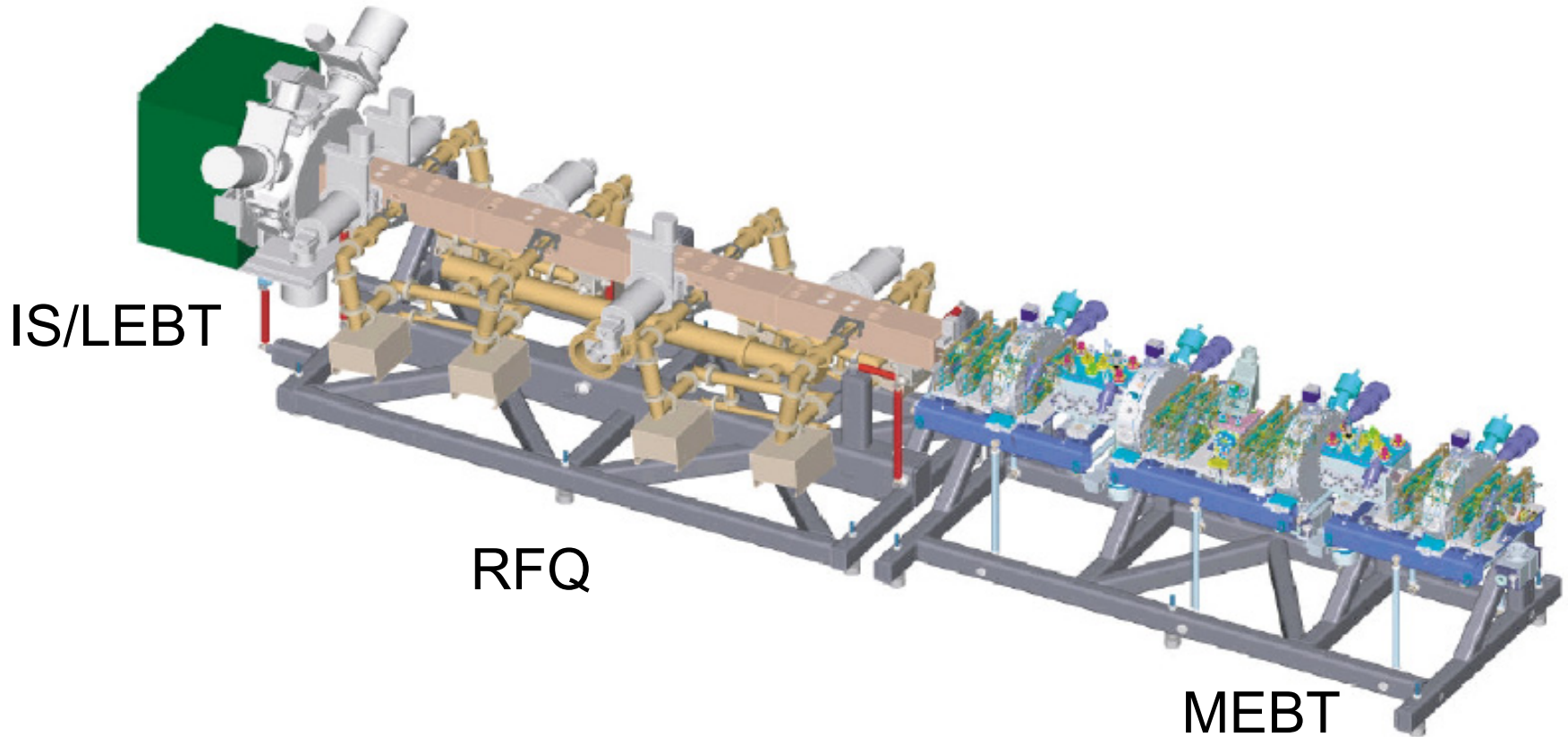
Parameters:

P beam on target	1.44MW	Duty factor	6%
I beam aver.	1.44mA	Rep. rate	60Hz
Beam energy	1 GeV	Pulse width	1ms

Front End Systems

1ms pulse

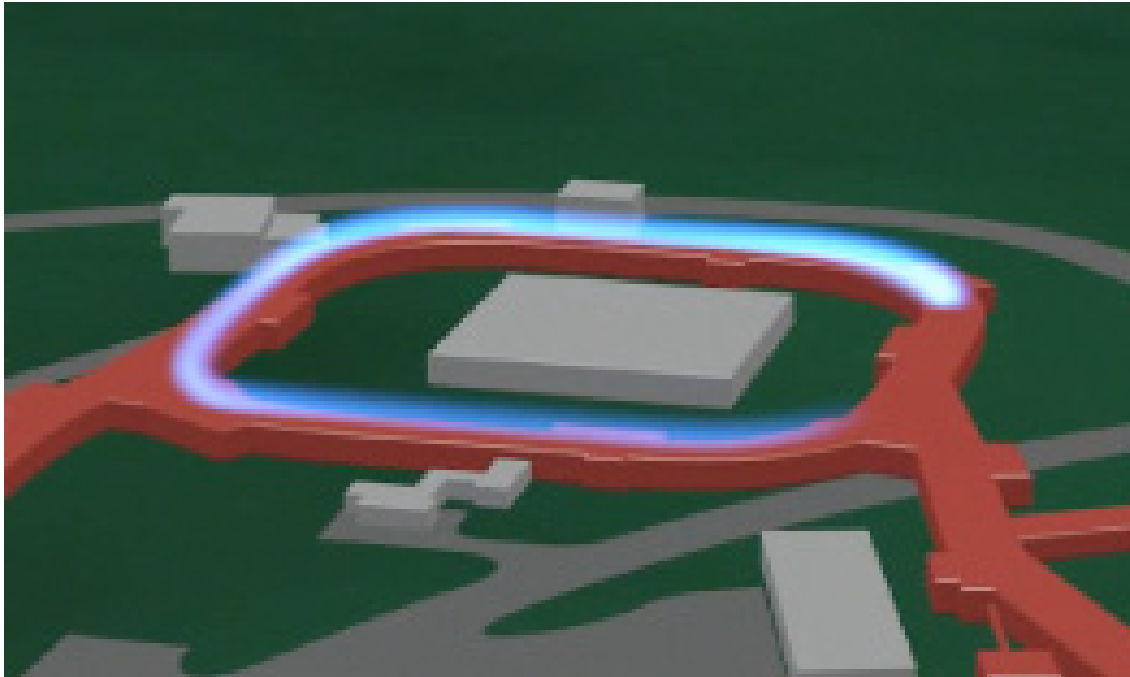
60Hz



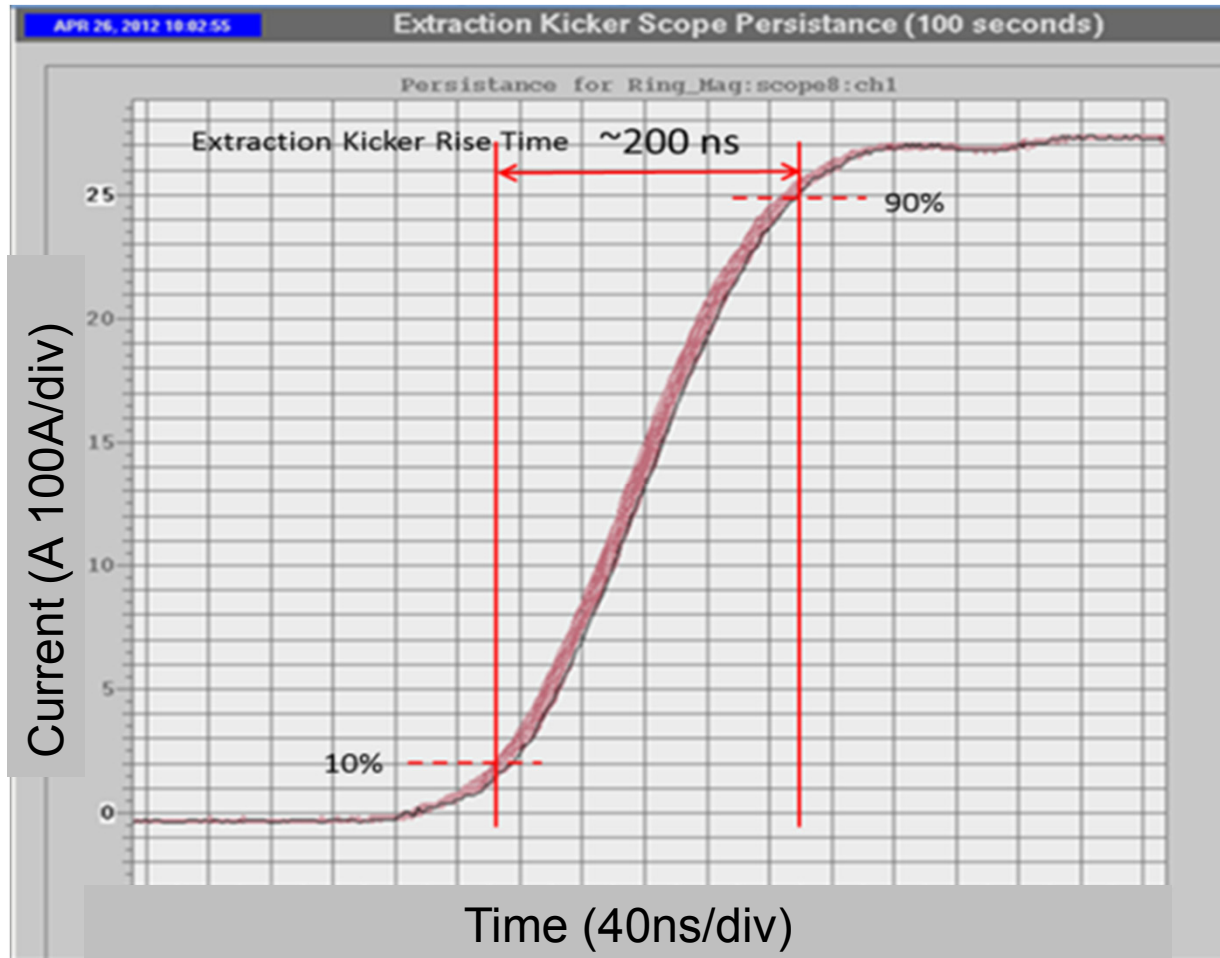
Alexander Aleksandrov

Chopper Timing Requirements

- Set by Accumulation Ring Beam Extraction Kicker Rise time
- Extraction kicker must turn on during gap
- Gap created by LEBT/MEBT chopping



Extraction Kicker #13 Magnet Current Rise Time



Pulse Timing

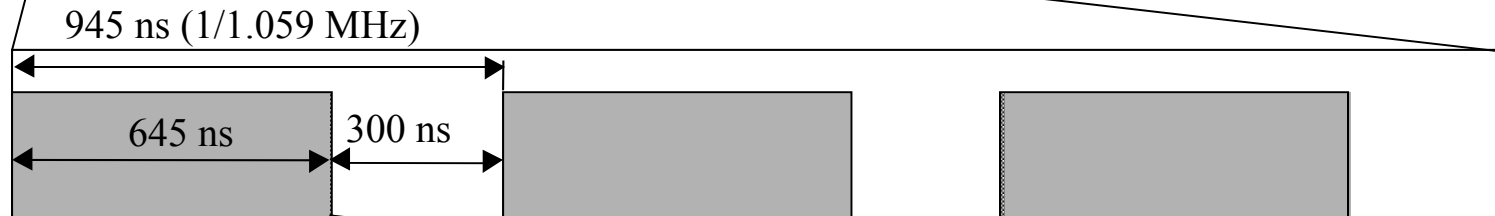
Macro-pulse

Structure
(made by the
Ion Source)



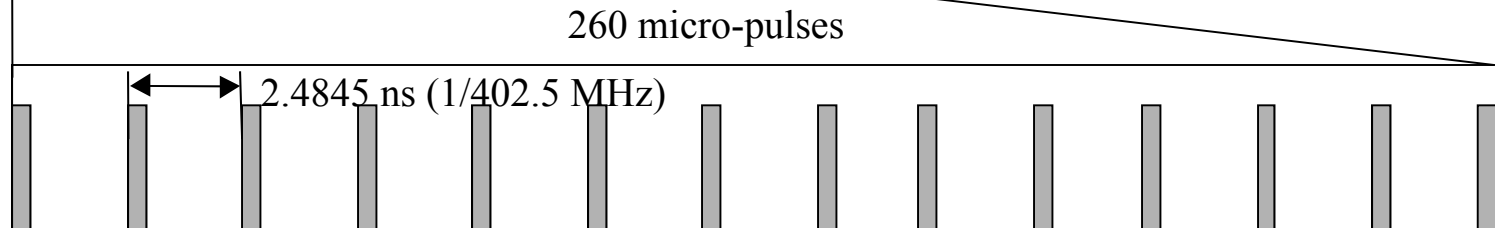
Mini-pulse

Structure
(made by the
choppers)



Micro-pulse

structure
(made by the
RFQ)



Alexander Aleksandrov

Pulse Timing

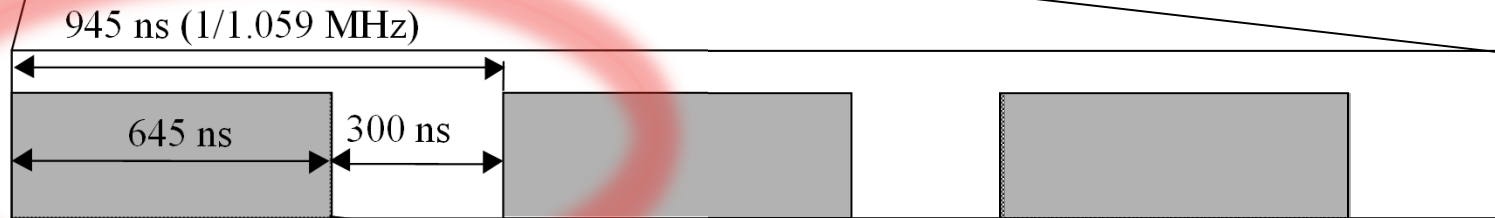
Macro-pulse

Structure
(made by the
Ion Source)



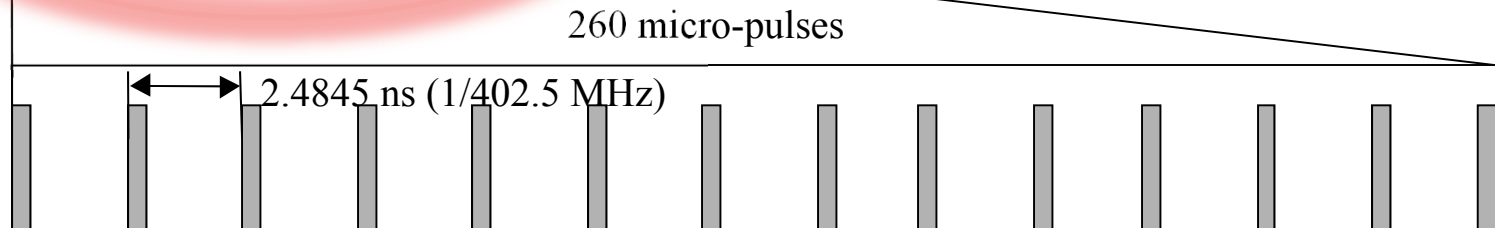
Mini-pulse

Structure
(made by the
choppers)



Micro-pulse

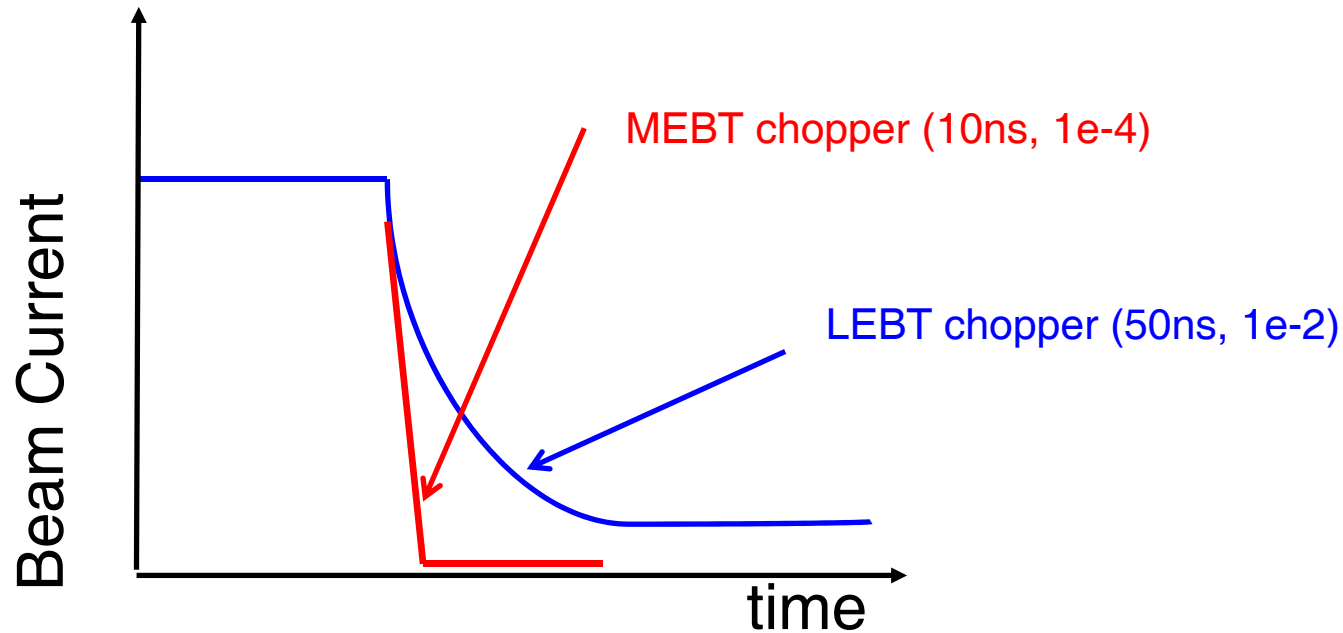
structure
(made by the
RFQ)



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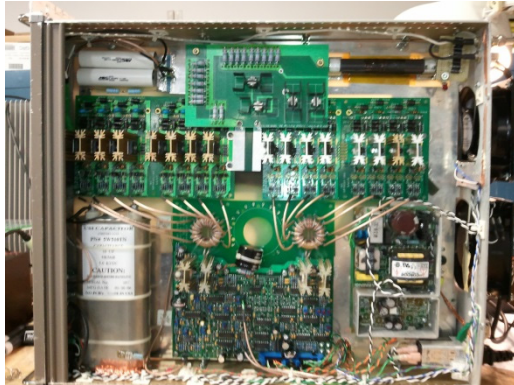
SNS Front End Chopping System

Two stage: LEBT + MEBT

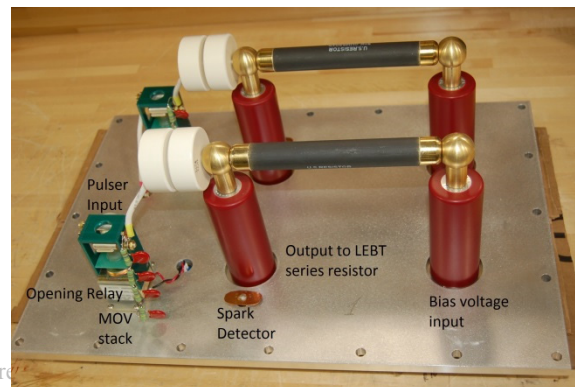
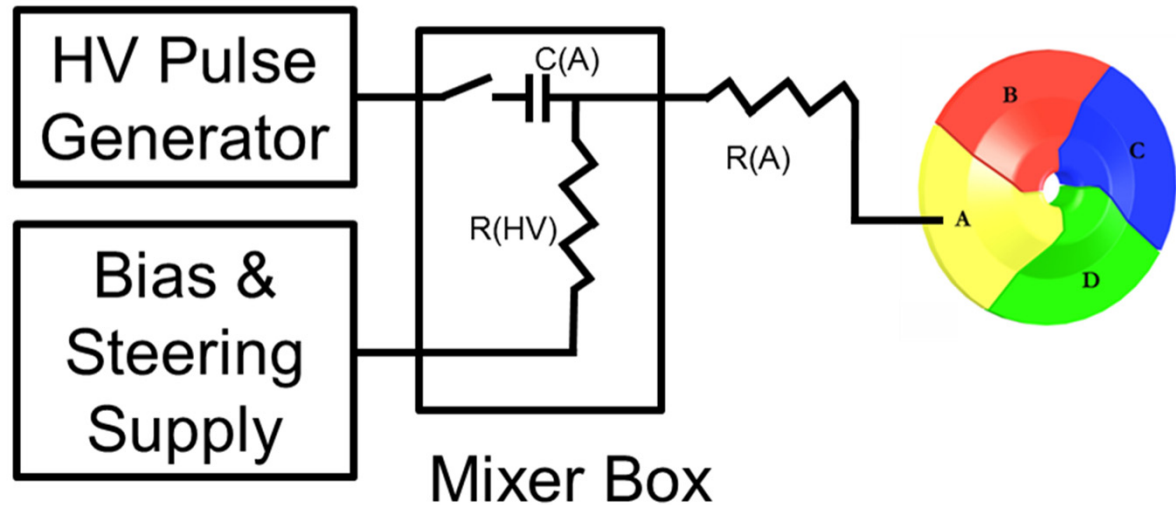


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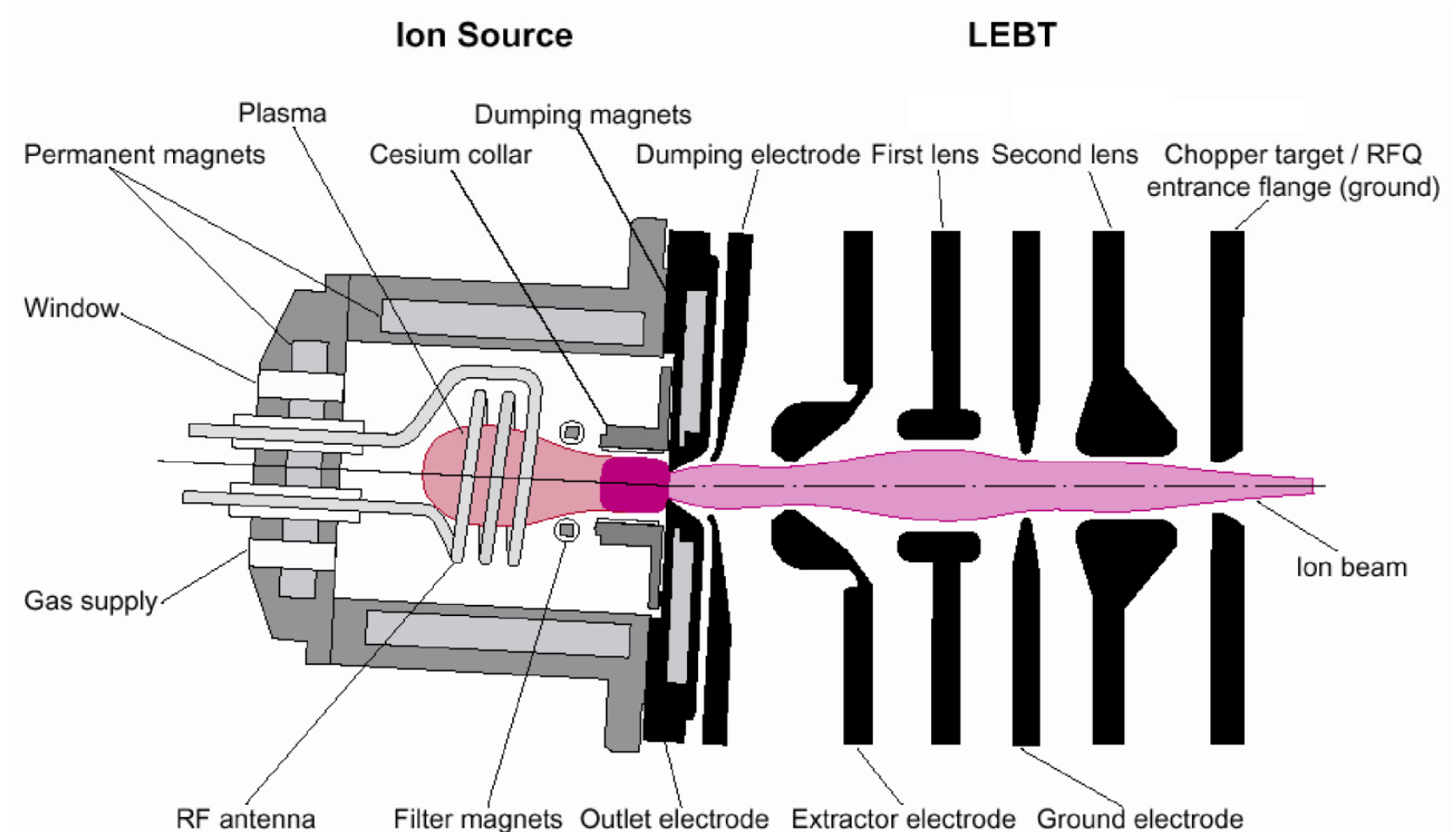
LEBT Chopper



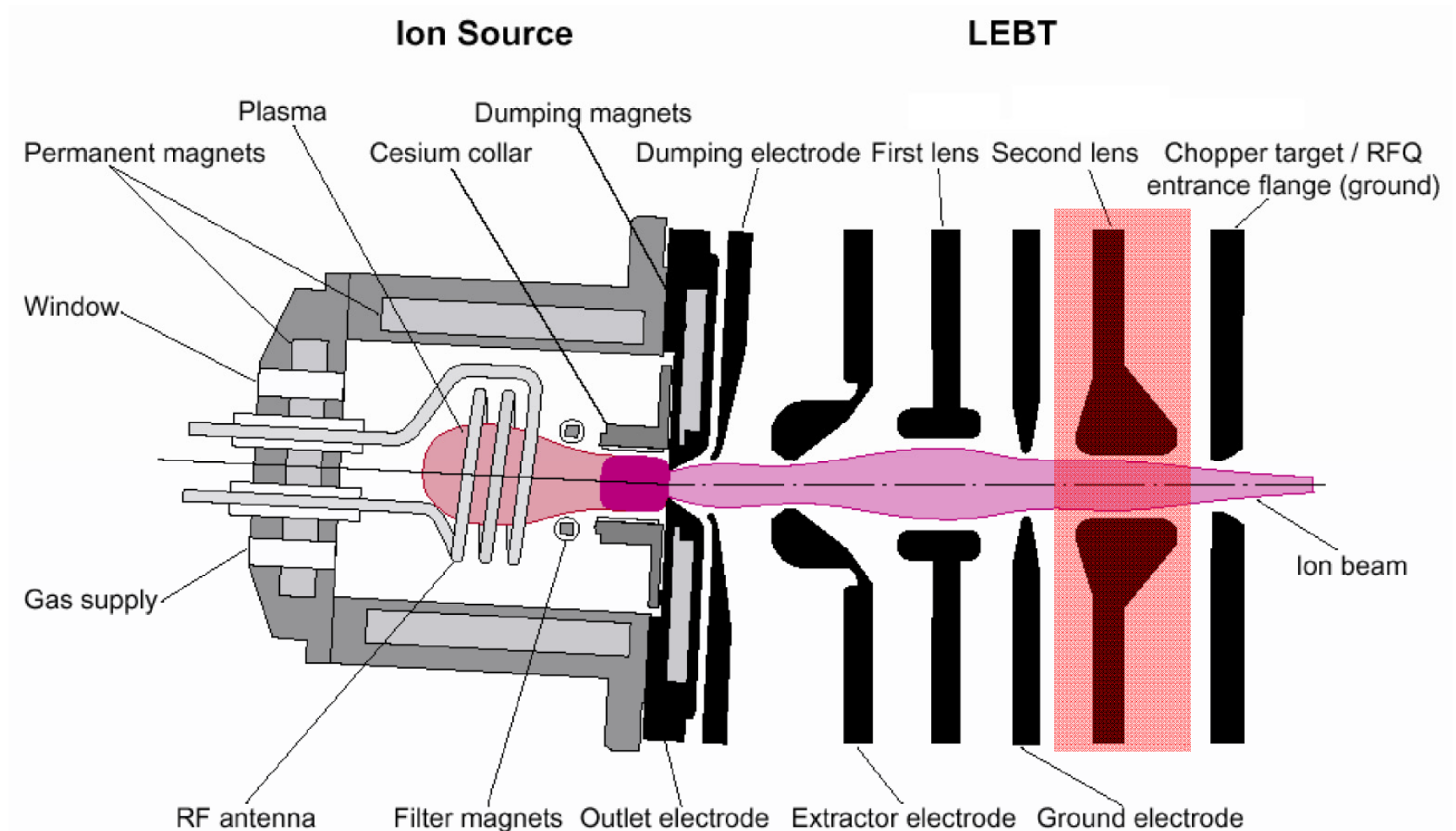
LEBT Chopper System (One Channel)



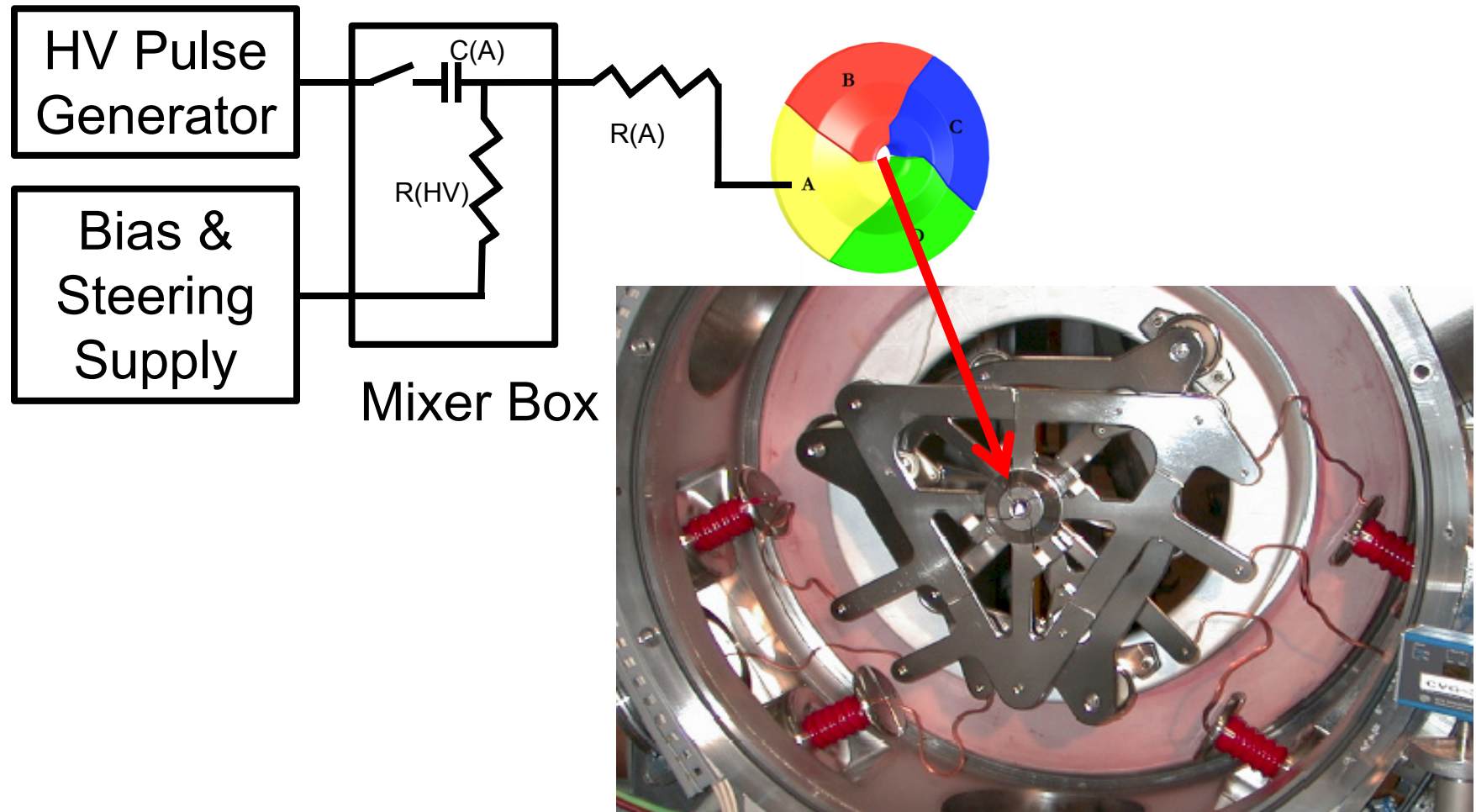
Ion Source and LEBT Second Lens



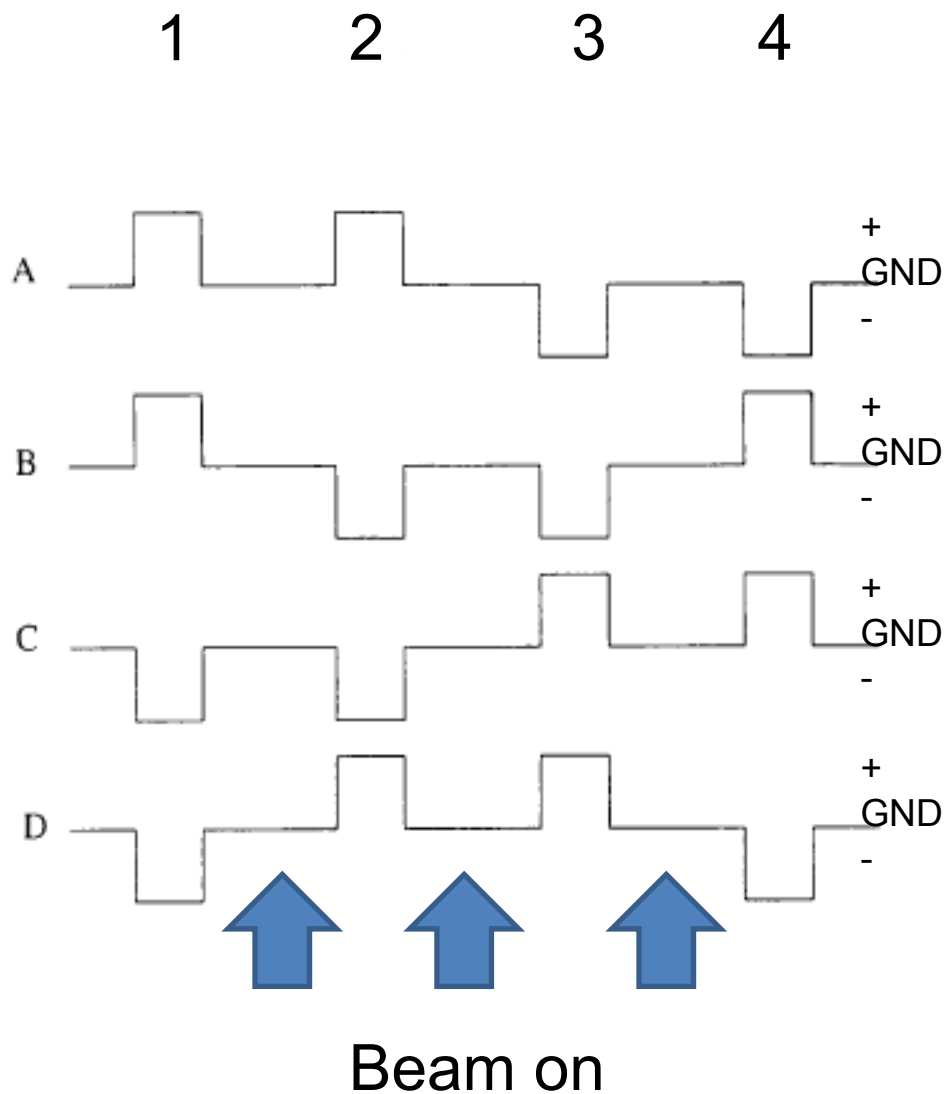
Ion Source and LEBT Second Lens



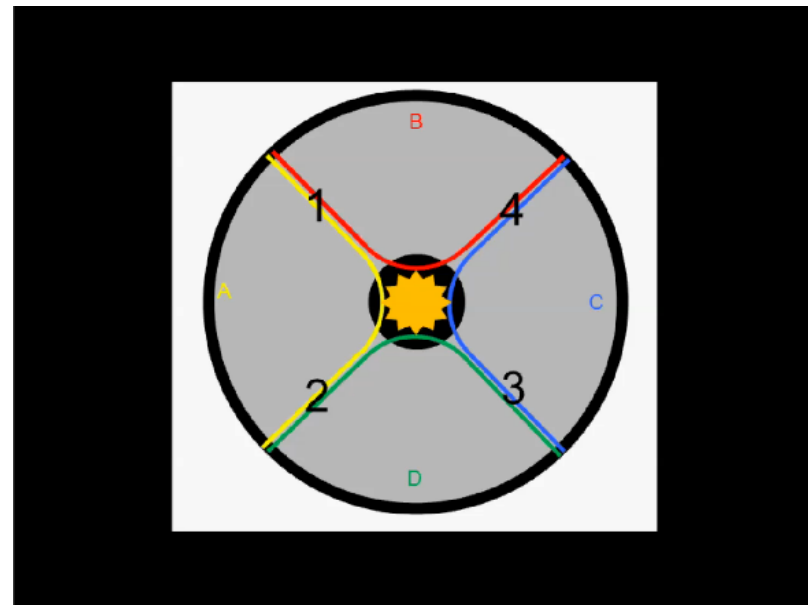
LEBT Chopper System (One Channel)



LEBT Four Phase Timing

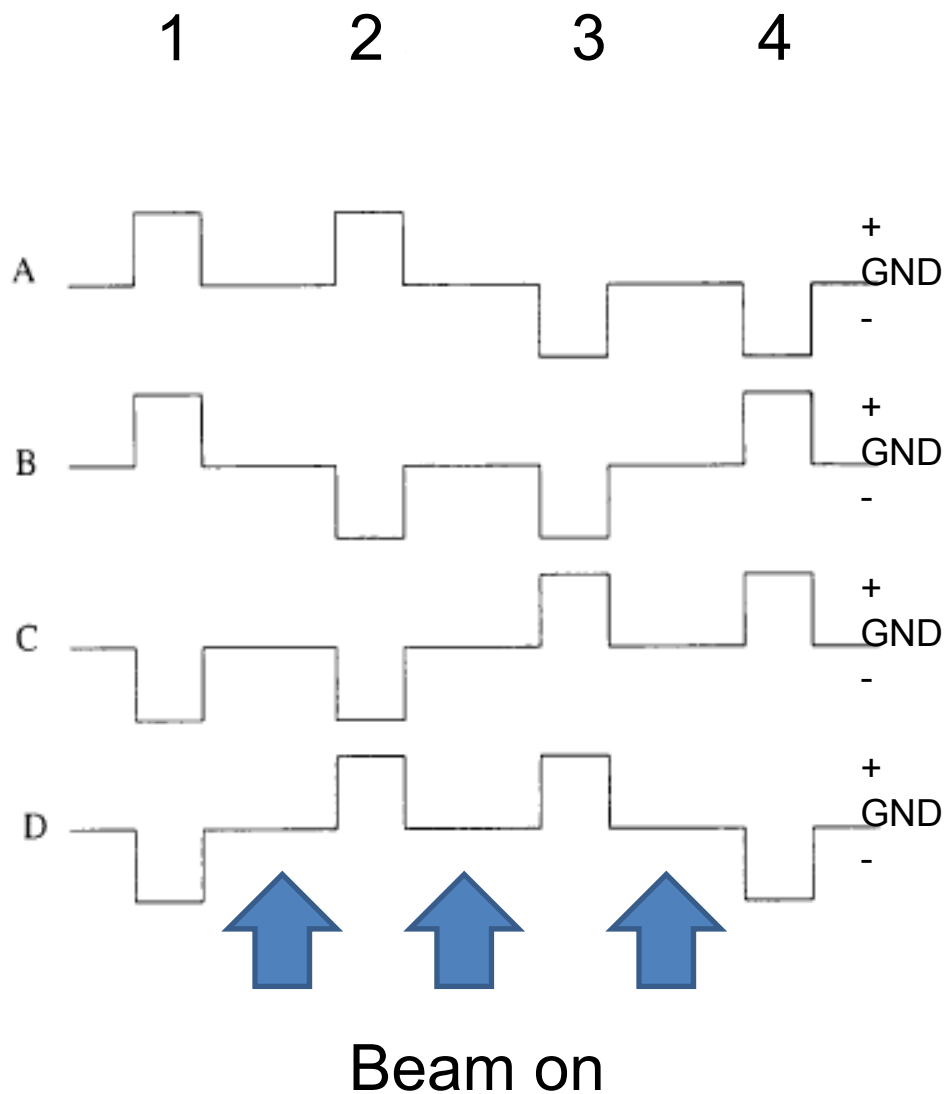


Diagnostic Plate

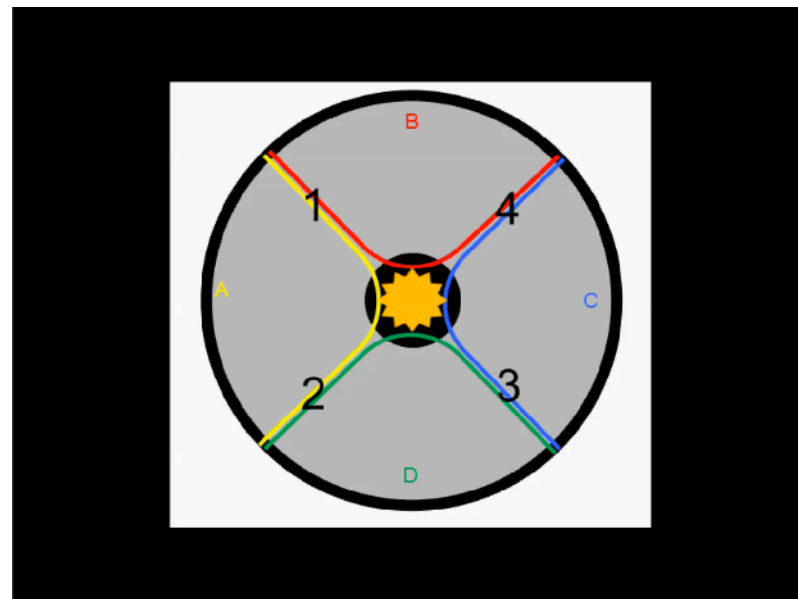


Normal Chopping Pattern

LEBT Four Phase Timing



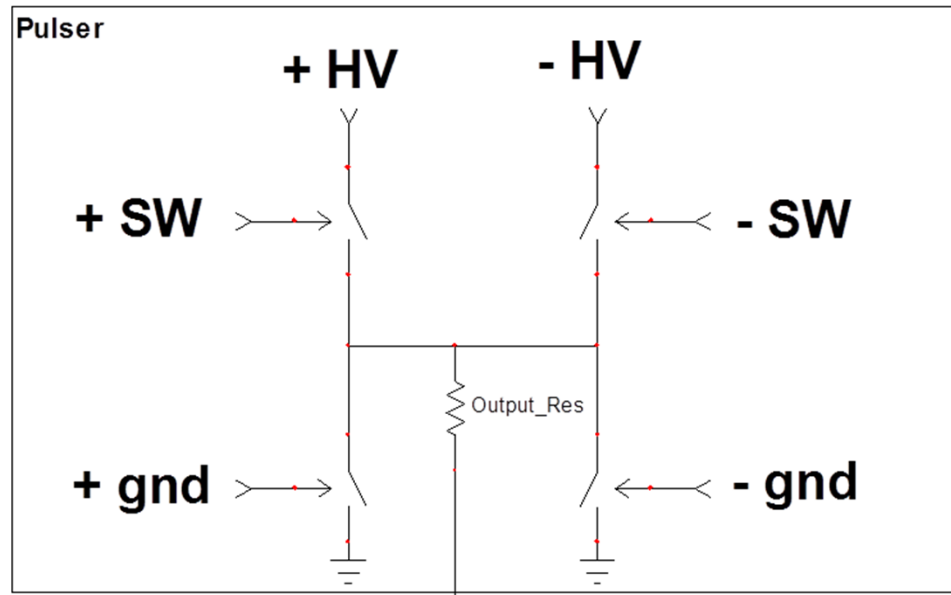
Diagnostic Plate



Beam Off Pattern

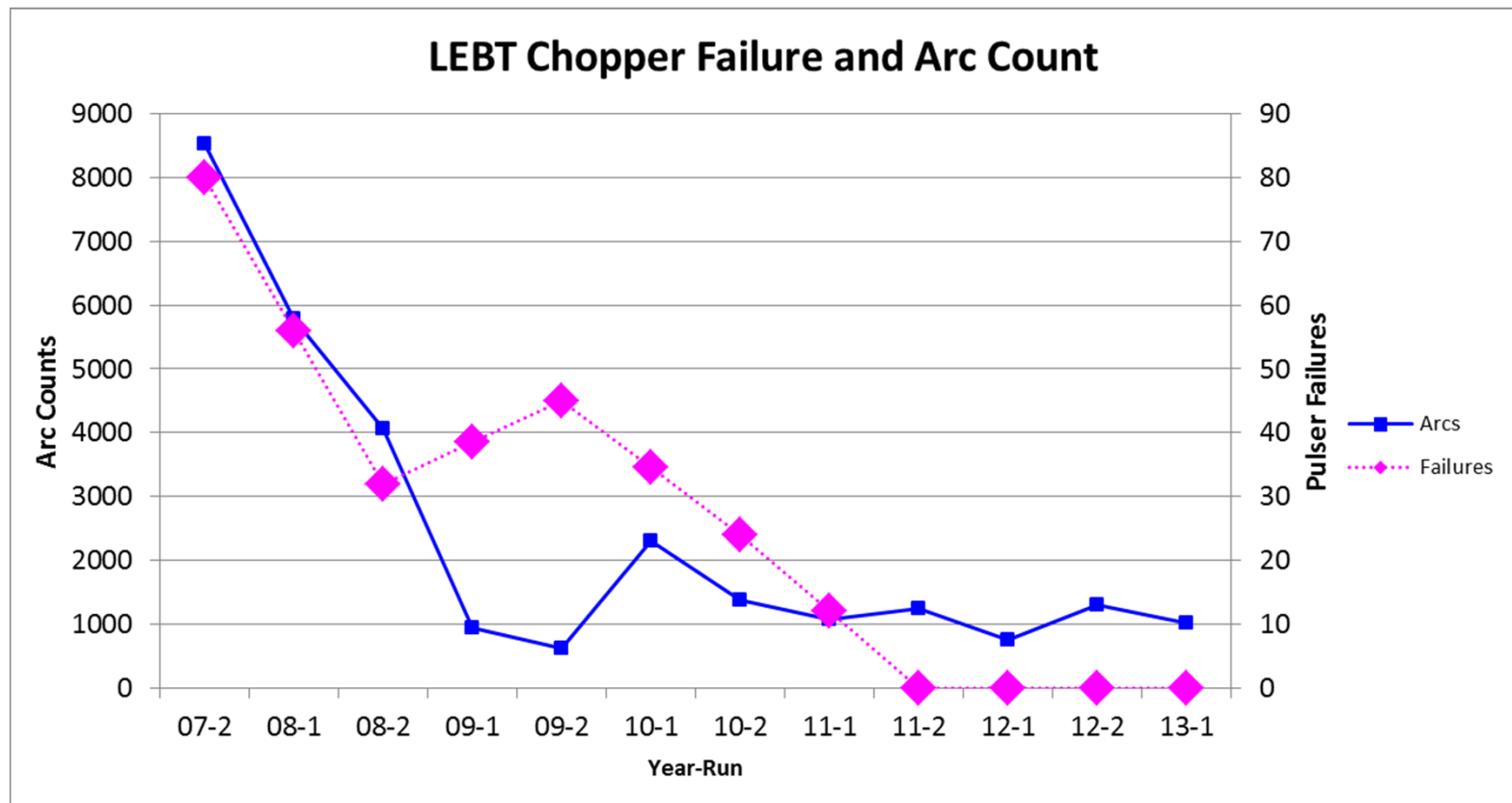
LEBT HV Pulse Generator Specifications

- Voltage +/- 3 kV
- Rise/ Fall Times < 50 ns on lens segment
- Variable pulse widths 200 ns –1000 ns
- Each switch is a stack of 8 MOSFETS



LEBT HV Pulse Generators Reliability

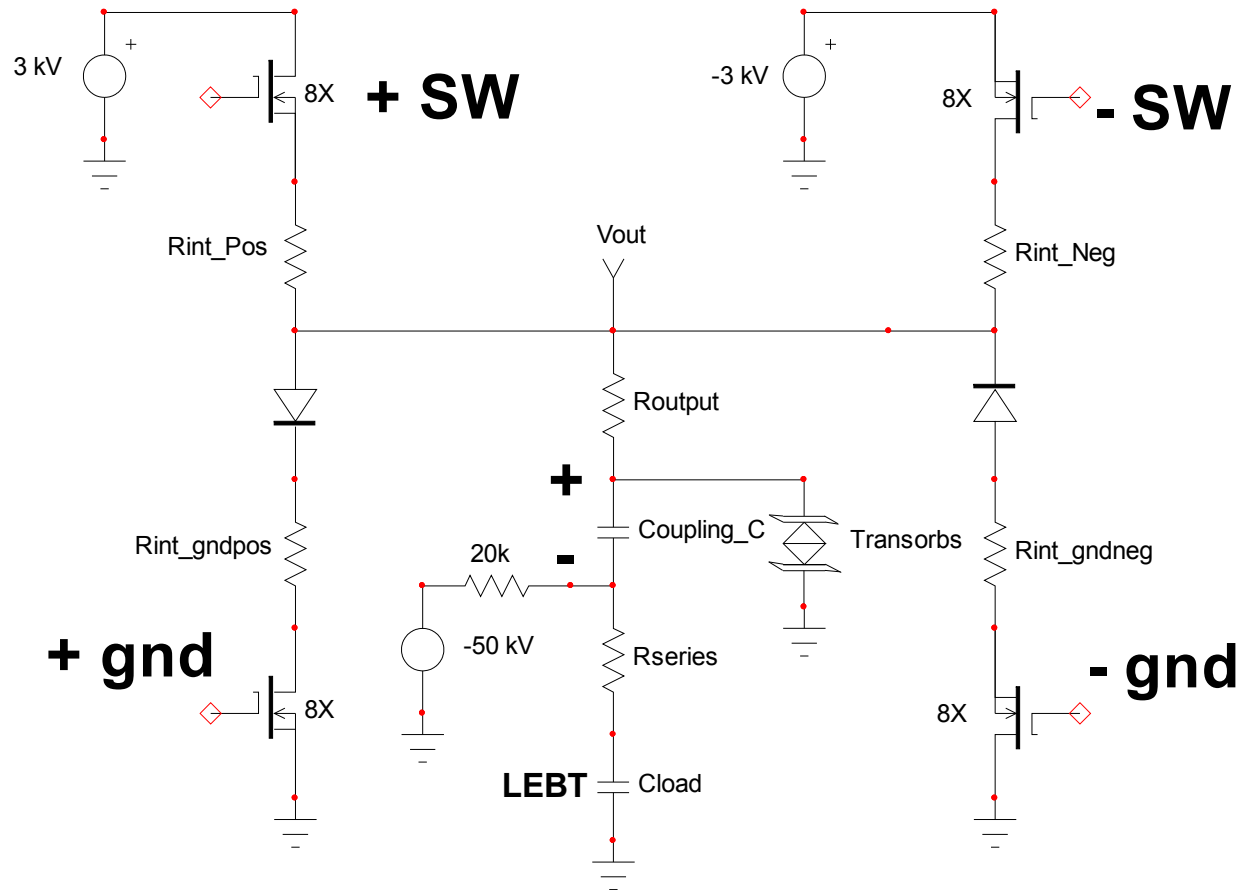
- Typically failures only with arcs
- ~30000 lens arcs since 2007
- Failure rate per arc low $<1\%$



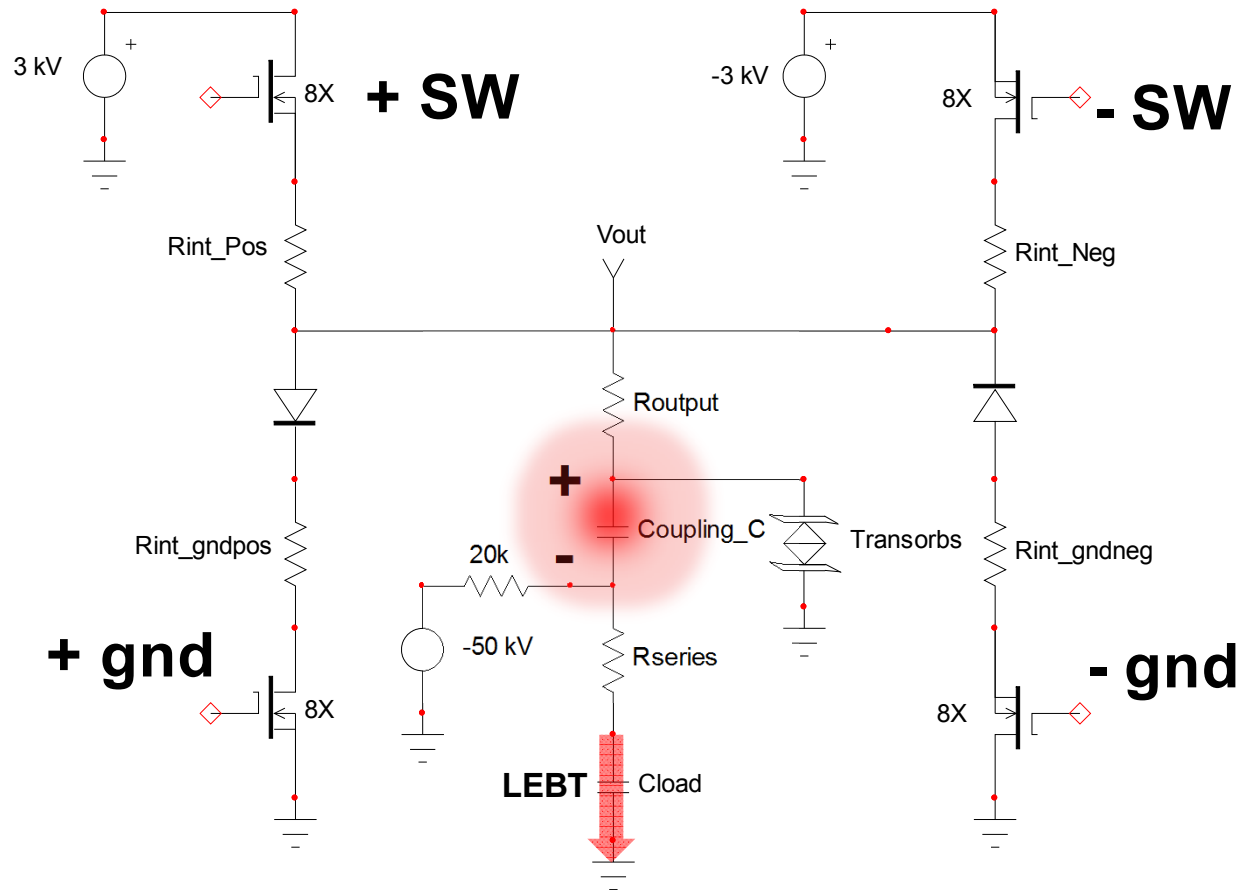
LEBT HV Pulse Generator Failure Modes

- HV MOSFET failure
- Initially suspected to be:
 - Over Current due to shoot through
 - Over Voltage of switch stack due to 50 kV arc
- Determined to be Over Voltage
- Attempted solutions
 - Raising series resistance (sacrificed rise/ fall times)
 - Adding MOVs
 - Graded voltage sharing of MOSFETs in stack
 - Adding HV diodes in series to share voltage hold off

Arc Event Failure Analysis

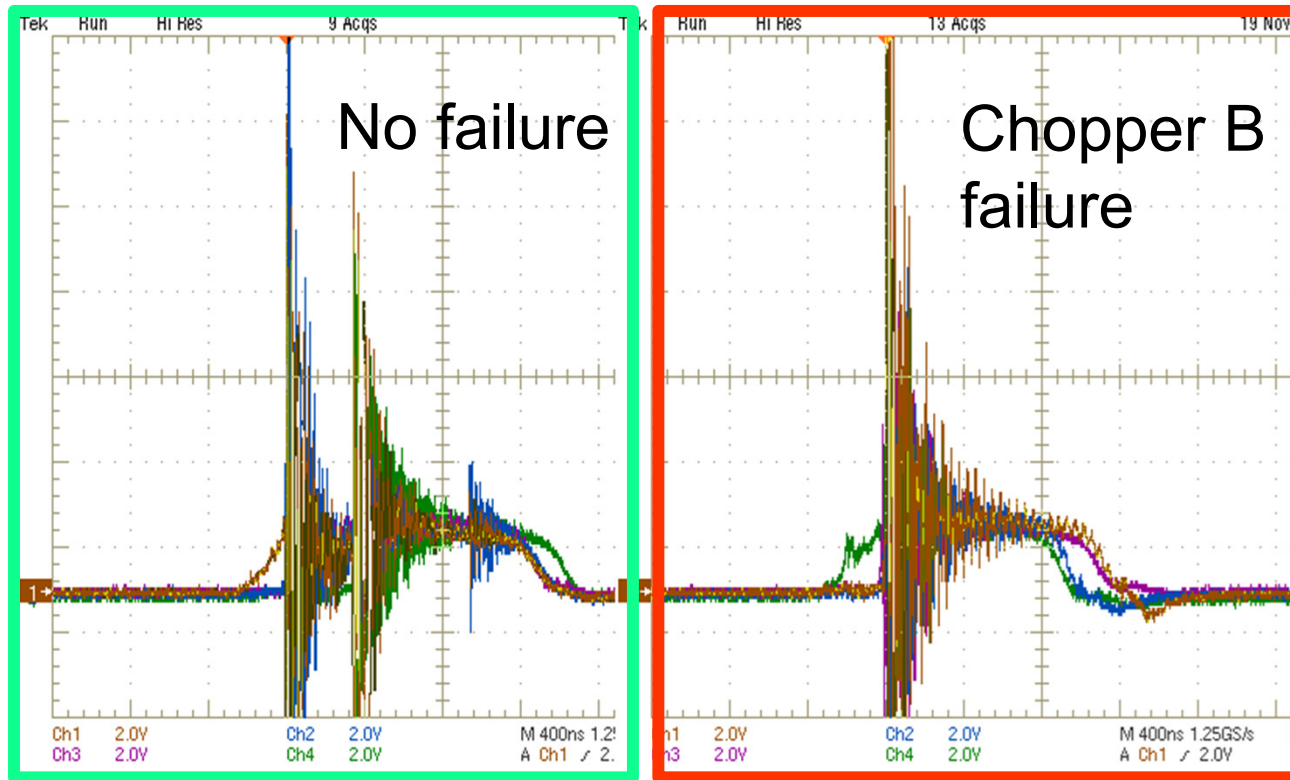


Arc Event Failure Analysis



Arc Events

Arc waveforms during OFF time

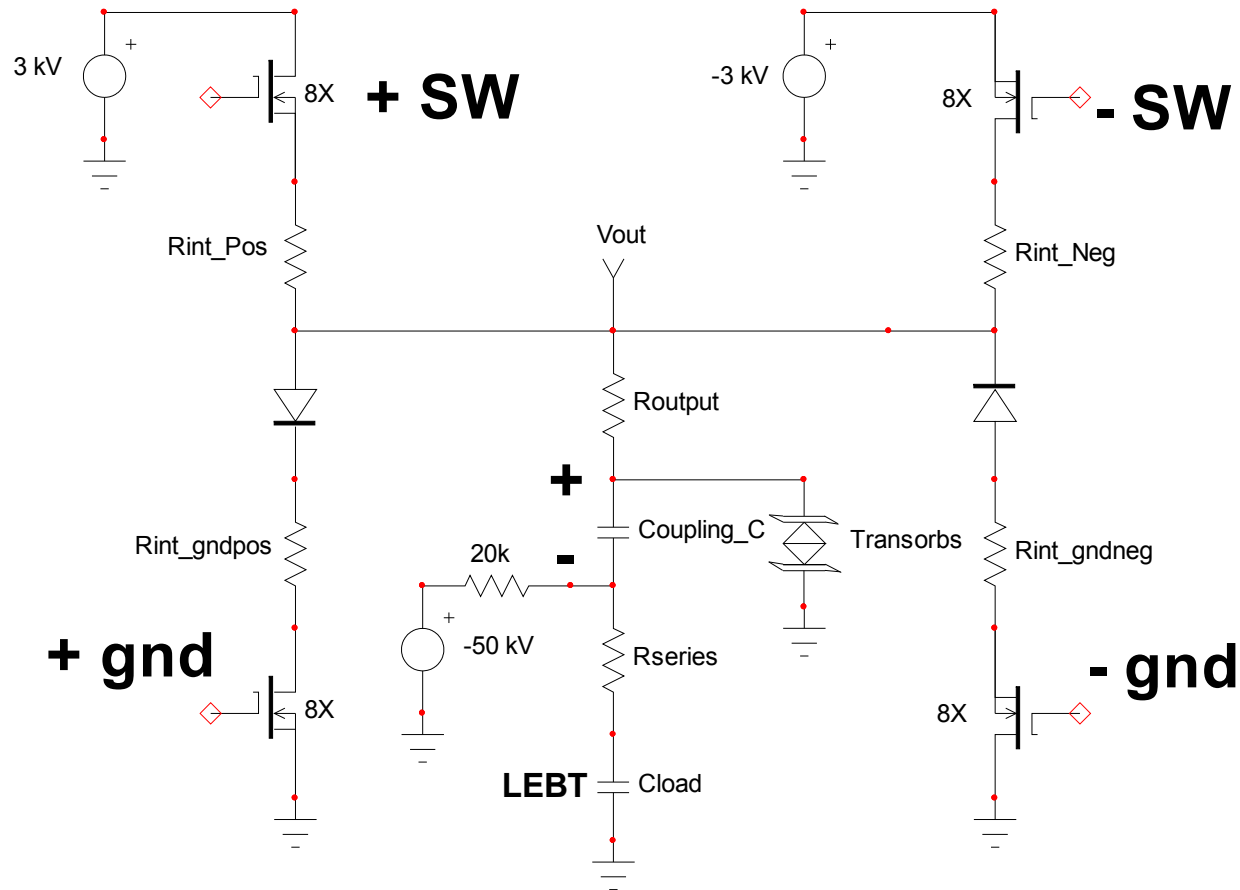


Ch1 = A, Ch2 = B, Ch3 = C, Ch4 = D

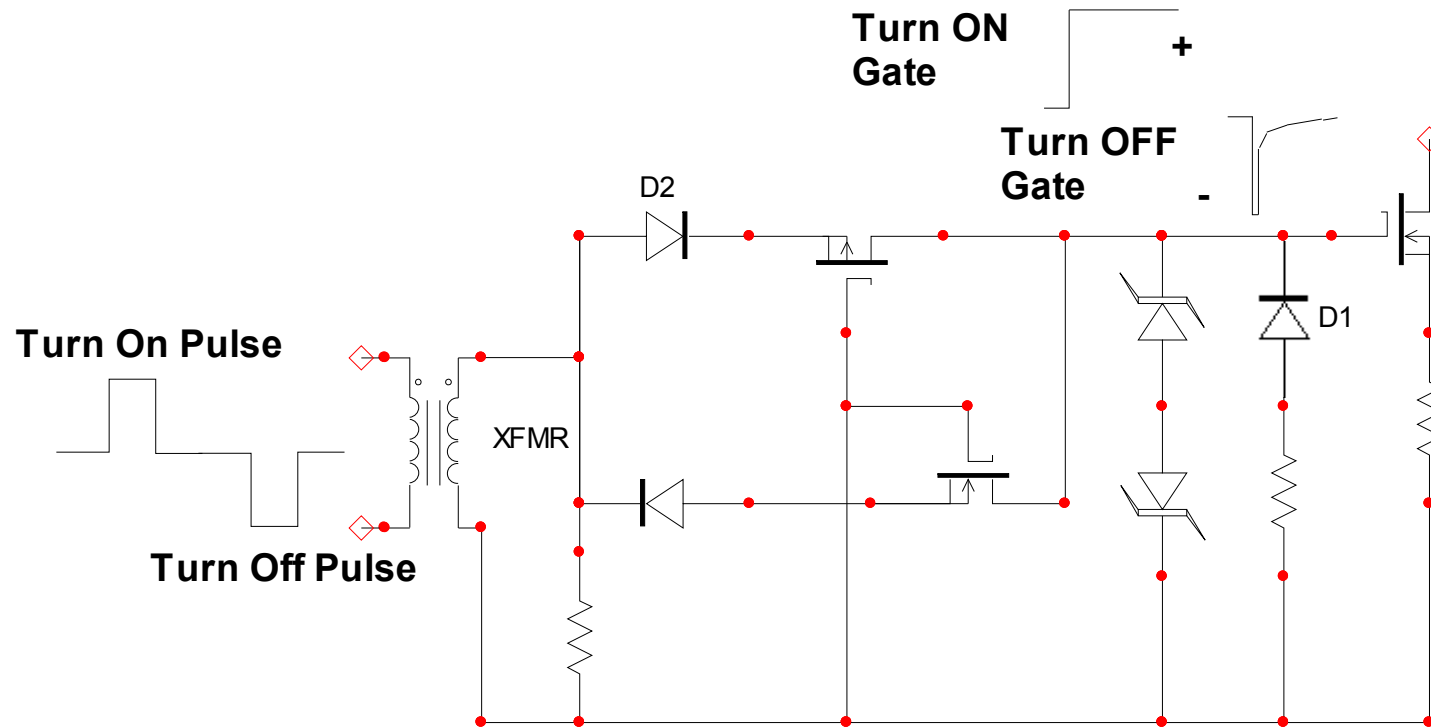
Chopper B Dead after this I

- No significant difference between arc event that doesn't damage pulse generator and one that does

Arc Event Failure Analysis – Gate Drive

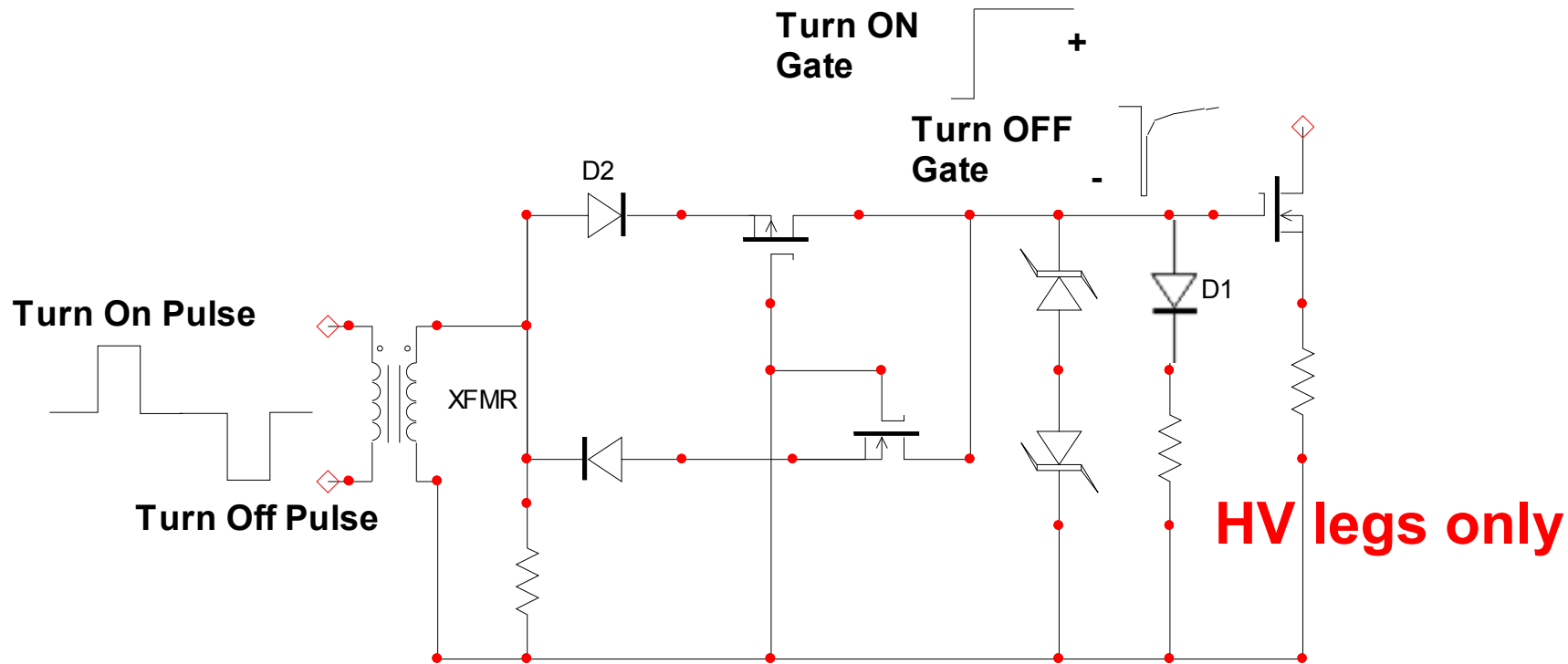


Power MOSFET Gate Drive Circuit



Positive gate pulses latched ON
Negative gate pulses decay to 0V after 4 μ s

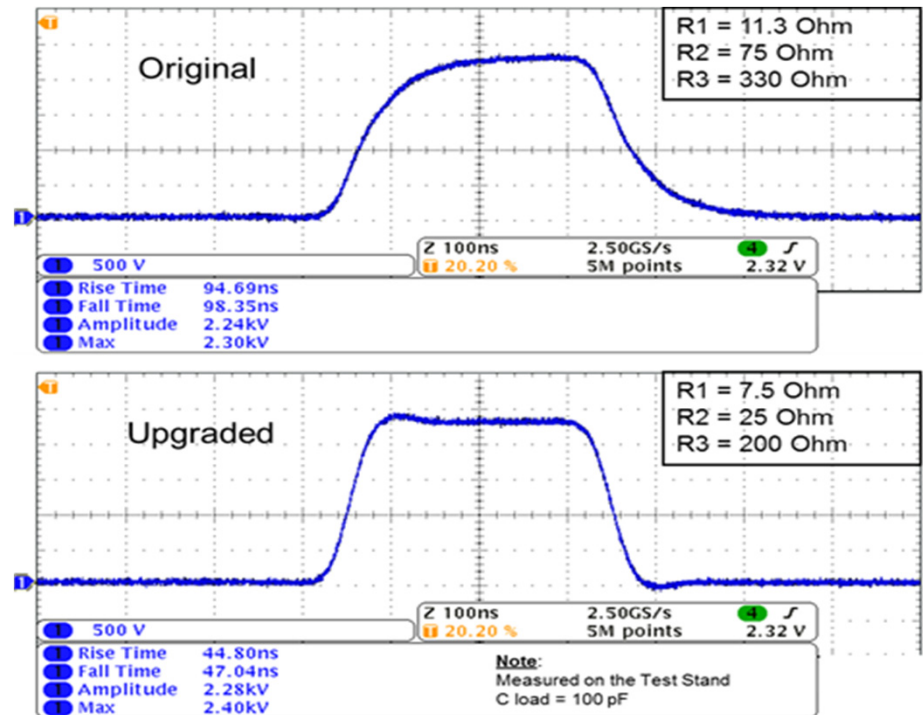
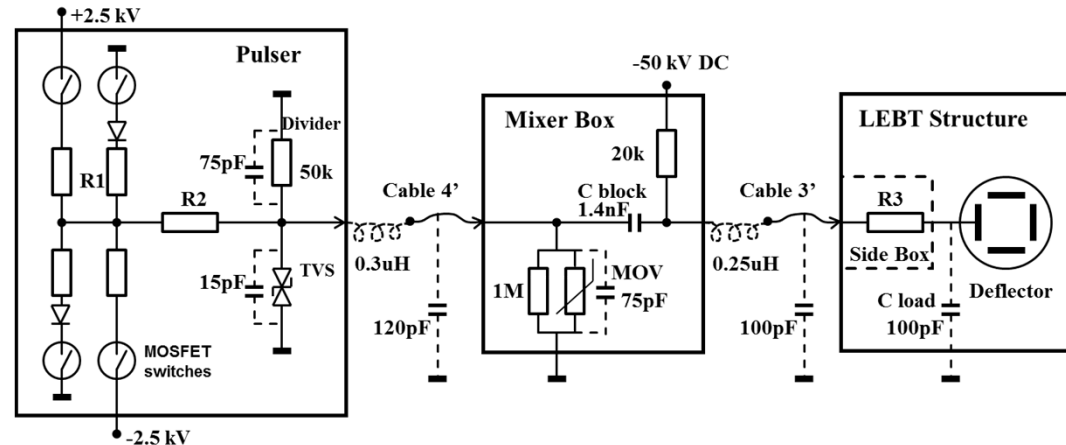
Power MOSFET Gate Drive Circuit



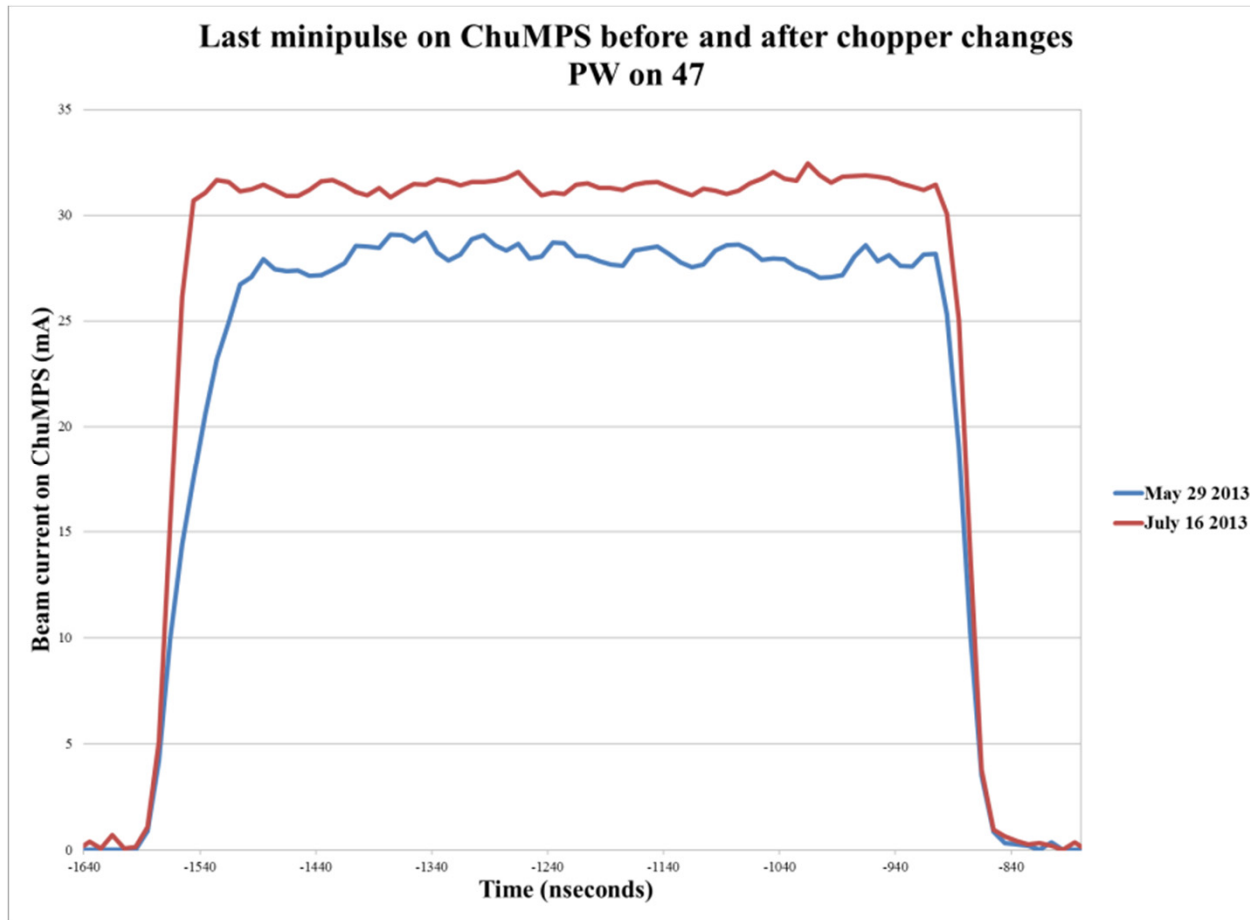
Positive gate pulses latched ON
Negative gate pulses decay to 0V after 4 μ s

LEBT Chopper Rise/ Fall Time

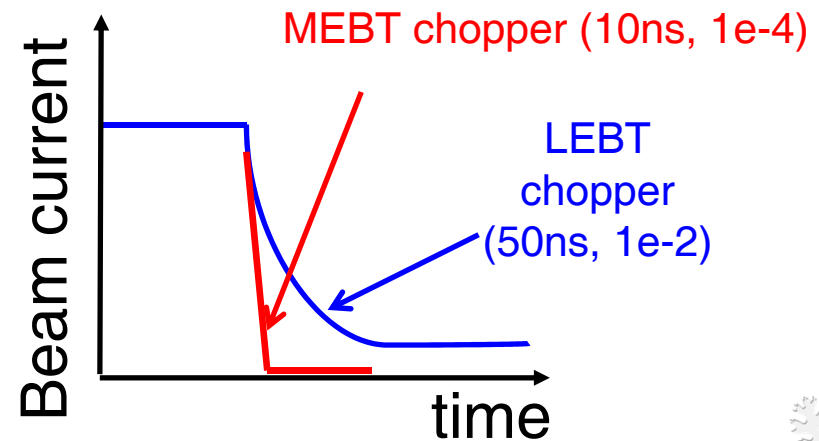
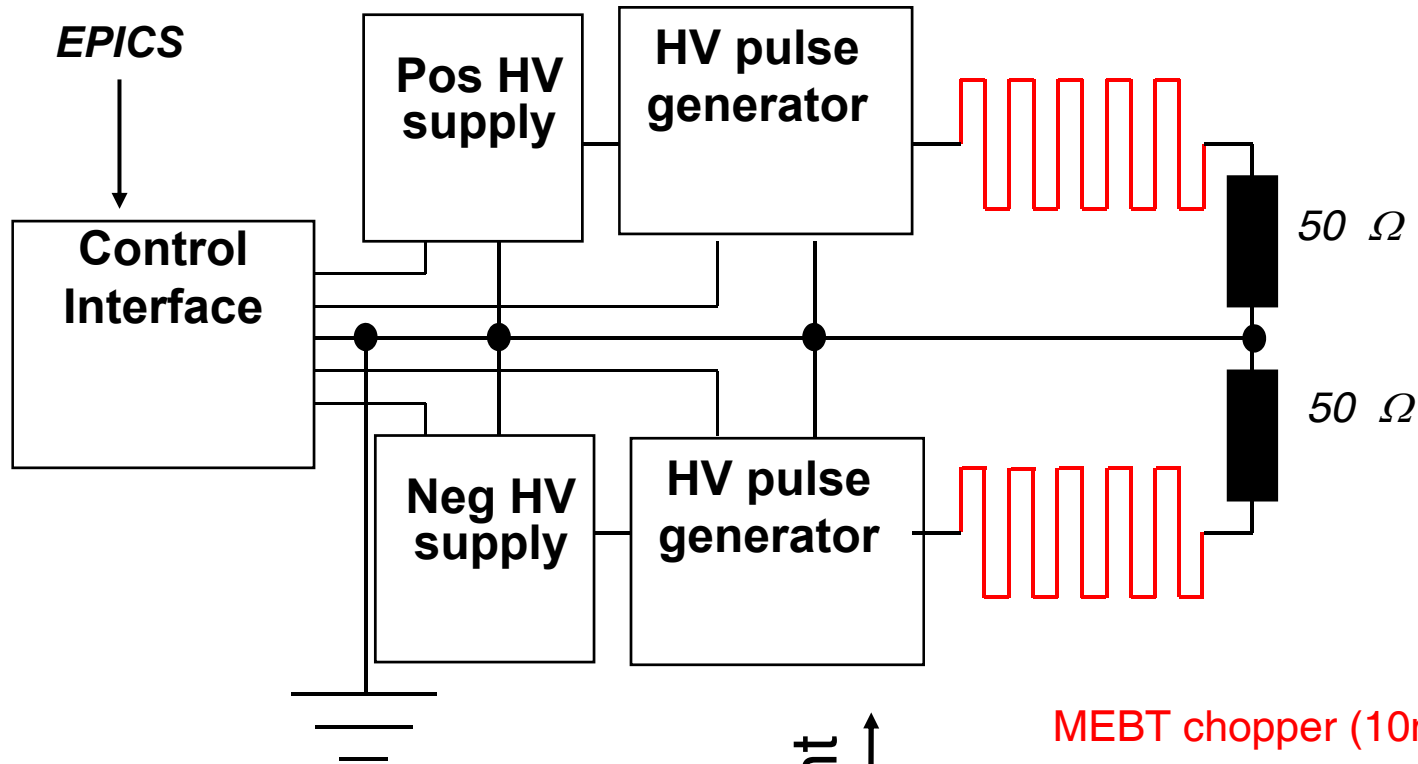
- Set by series resistance and capacitance
- Voltage division ratio kept constant
- Pulse rise and fall time reduction
 - 50% reduction achieved
 - Meets original specification



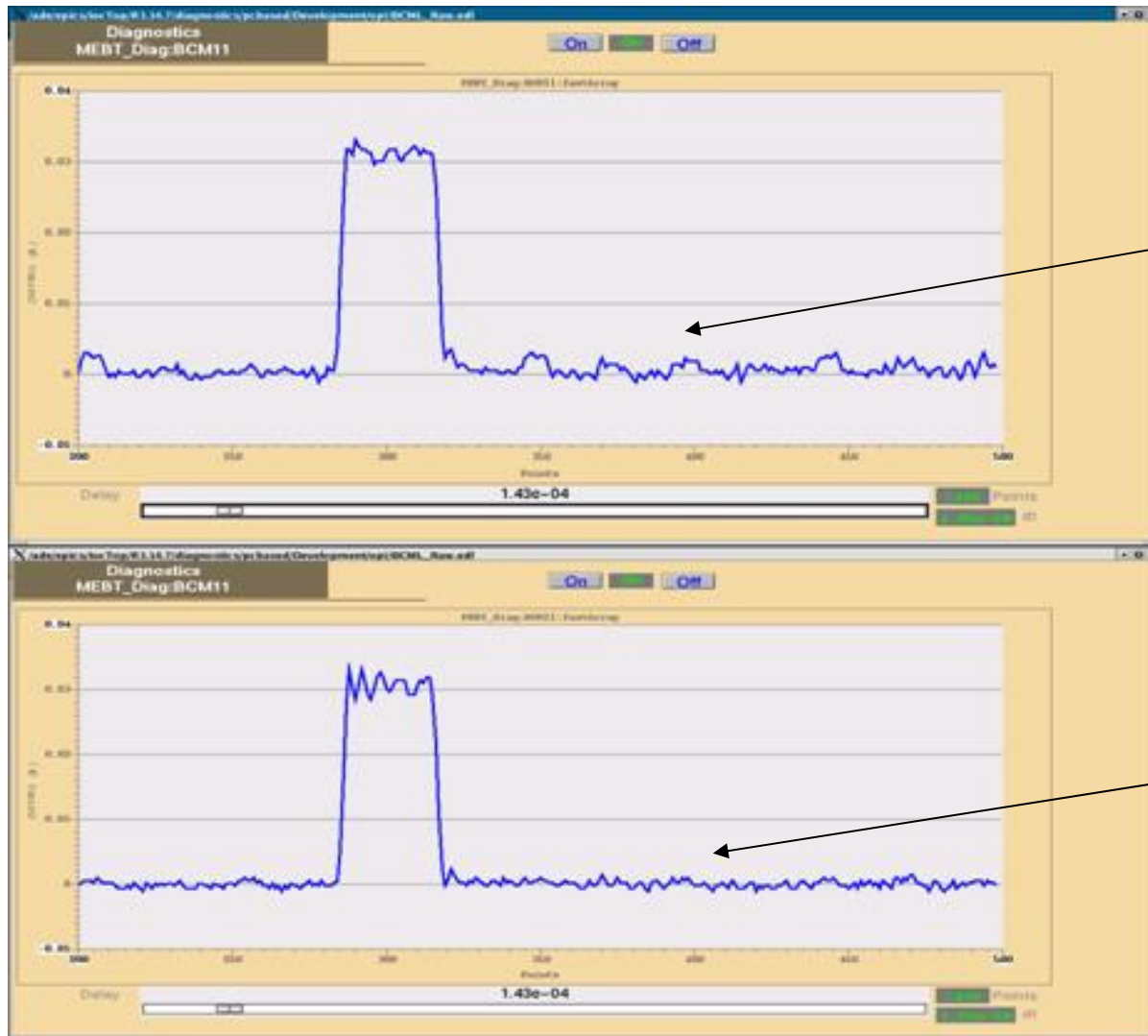
LEBT Chopper Rise and Fall Time as seen by the Beam



MEBT Chopper



MEBT Chopper assisting beam clearing



Beam current
measured with
LEBT chopper
alone

Beam current
measured with
LEBT and
MEBT
choppers

MEBT Structure Problems

- **Reliability of the deflector structure**
 - **Water Leaks**
 - **High voltage breakdown**
 - **Thermal design was not adequate**
 - **Both structures delaminated due to overheating even when operated with <1% of design beam power**
 - **Protection from beam striking the structure is not adequate**

MEBT Structure

Original “meander line”

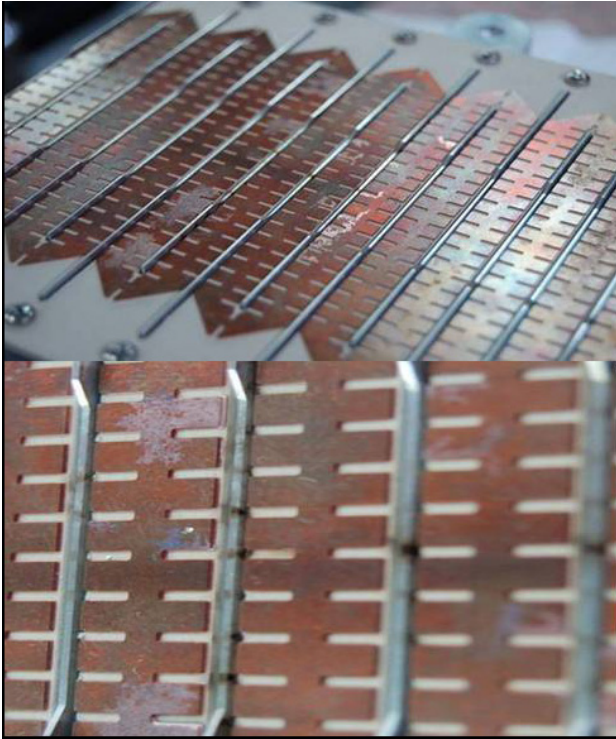
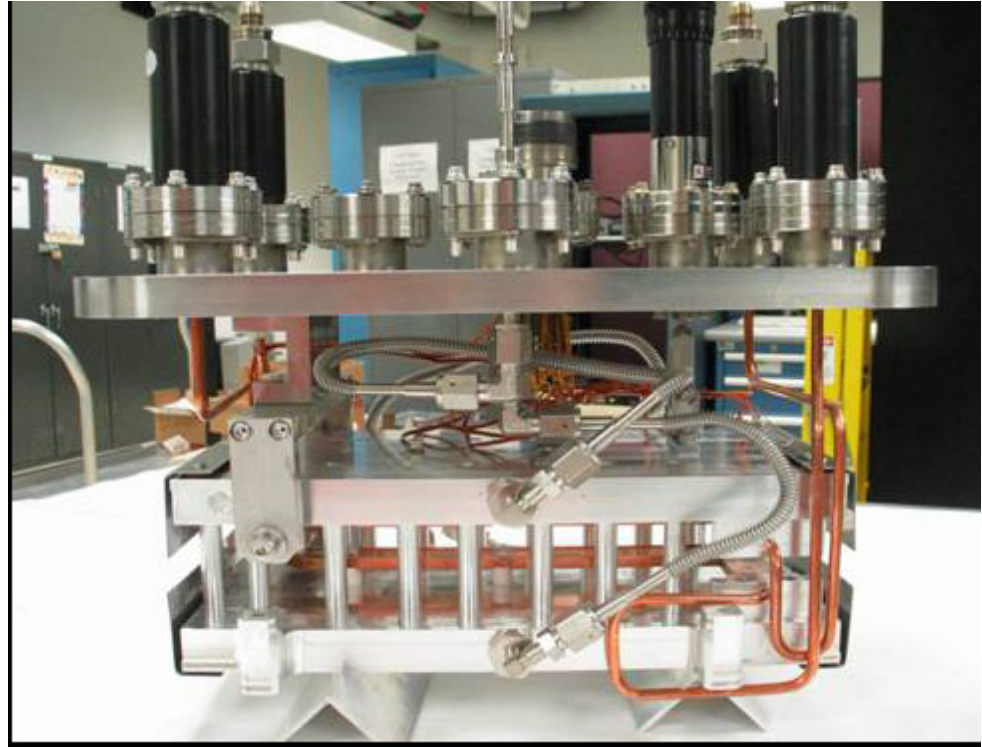


Photo of the damaged PCB structure of the original chopper deflector.

New “parallel plate”



New MEBT chopper deflector

“NEW DESIGN OF THE SNS MEBT CHOPPER DEFLECTOR”

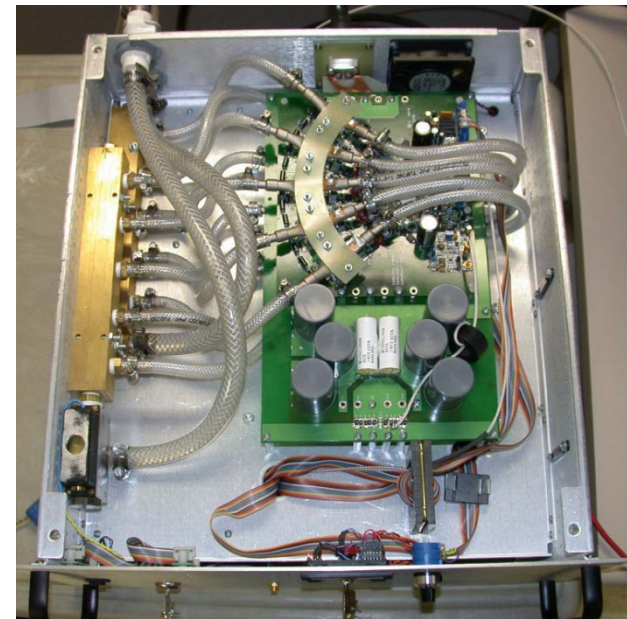
A. Aleksandrov, C. Deibele, T. Roseberry.

Oak Ridge National Laboratory, Oak Ridge, TN 37830 USA

Proceedings of PAC07, pp. 1817-1819, Albuquerque, New Mexico, USA

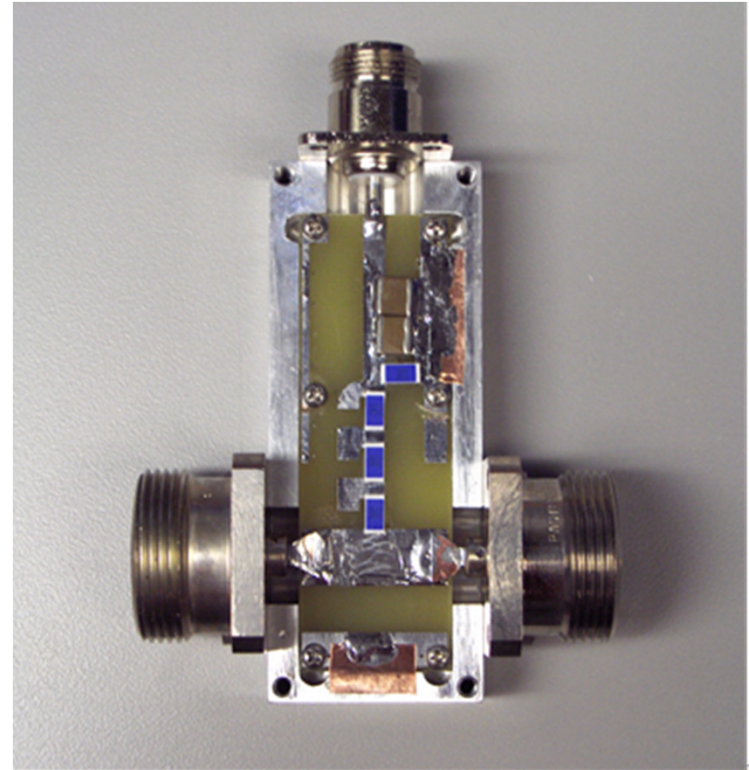
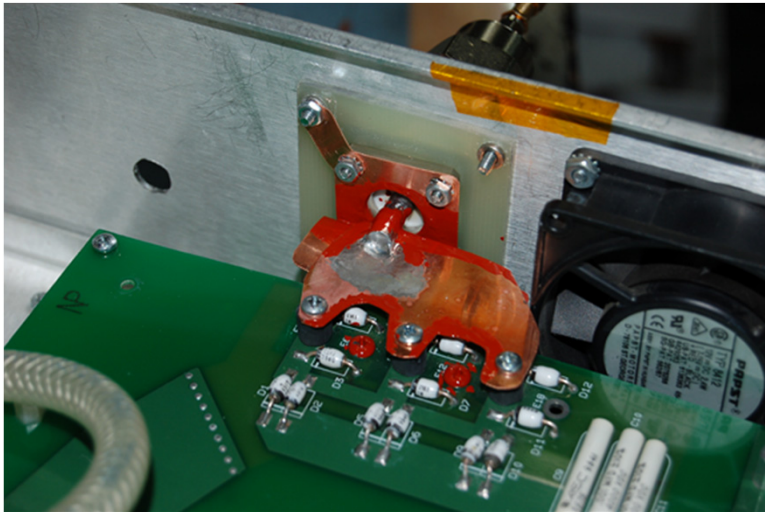
High Voltage Pulse Generator Design

- Stack of 5 high speed MOSFETS in series as HV switch
- Single Polarity with active grounding switches
- Water cooled



MEBT Pulse Generator Improvements

- Output Impedance matching to reduce edge reflections
- Microstrip high impedance probe for fast rise/ fall time measurements



Summary

- **Increased Availability of LEBT and MEBT Chopper Systems**
- **Improved arc event survivability of the LEBT Chopper**
- **Improved Rise and Fall times of the LEBT Chopper**
- **Improved reliability of the MEBT Structure**
- **Impedance matching of MEBT Pulse Generators**
- **Improved Cooling of MEBT Pulse Generators**
- **Future work**
 - **Redesign of LEBT Pulse Generators to improve control timing**
 - **Investigate using MEBT Pulse Generator as a fast beam abort**

Questions?